

A Real-Time COVID-19 Data Visualization and Information Repository in the Philippines

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Abstract—The current COVID-19 pandemic has caused undeniable damages in varying sectors throughout the world, from healthcare, to economy, to education and politics. One of the pressing issues that the spread of this disease has also caused is the corresponding propagation of misinformation and “fake news” on the coronavirus. In response to this, the Sustainable Development Solutions Network (SDSN) Youth Philippines has proudly created and seeks to expand a consolidated and comprehensive Interactive Philippine Dashboard (<https://covid19.sdsnyouthph.org/>), a one-stop shop for all relevant information on COVID-19, specifically designed to be understood by the youth.

Keywords—COVID-19; misinformation; data analysis; data visualization; health informatics; web application; youth-friendly

I. INTRODUCTION

On 31 December 2019, the World Health Organization (WHO) received reports of a cluster of pneumonia-like cases of unknown cause in Wuhan, Hubei, People’s Republic of China [1]. This was quickly determined to be caused by a novel coronavirus which was called as severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). The outbreak was declared a Public Health Emergency of International Concern on 30 January 2020 where the Philippines also reported its first case. The Philippines reported its first COVID-19-related death on 1 February 2020 [1]. By 11 February 2020, WHO announced a name for the new coronavirus disease: COVID-19 [2]. The outbreak has since infected almost 200 countries and more than 73 million cases according to John Hopkins University [3].

The current COVID-19 pandemic has caused undeniable damages in varying sectors throughout the world, from healthcare, to economy, to education and politics. One of the pressing issues that the spread of this disease has also caused is the corresponding propagation of misinformation and “fake news” on the coronavirus. It is often difficult for people to find streamlined sources of information that are accurate and of high quality as well. As such, most people, especially young people, will turn to more convenient, albeit less accurate, sources, instead.

In response to this, the Sustainable Development Solutions Network (SDSN) Youth Philippines has proudly created and seeks to expand a consolidated and comprehensive Interactive Philippine Dashboard (<https://covid19.sdsnyouthph.org/>), a one-stop shop for all relevant information on COVID-19, specifically designed to be understood by the youth.

Launched in 2015, SDSN Youth Philippines is the global youth division of the Sustainable Development Solutions Network (SDSN), an initiative to amplify the tremendous energy and capabilities of youth in generating broader buy-in for the Sustainable Development Goals (SDGs). Recently, SDSN Youth was also included in the official United Nations Task Force on COVID-19 Response, with an institutional mandate to create solutions that respond to the current pandemic. In response to the current pandemic, the Philippine team was recently appointed to spearhead the regional projects for Asia for SDSN Youth’s COVID-19 efforts [4].

Information included on the dashboard includes the following features: (1) real-time updates and graphs on the current cases, deaths, and recoveries as well as testing data in the country, (2) a consolidated list of news from local and international news organizations, (3) information such as symptoms, FAQs, parenting tips and even home care tips, (4) live video feed of government press briefings, (5) myth-busting, fact-checking, and feature articles on COVID-19, (6) COVID-19-related donation and volunteering opportunities, (7) good news and youth-led initiatives, and (8) Demographic Vulnerability Map. The information on the platform has been carefully curated in line with the needs of young Filipinos and is in accordance with the UN Task Force’s approach and directives on COVID-19 response.

The primary objective of this study is to present the framework for the interactive data visualization, analysis, and information repository, how the data is gathered and analyzed, and how the data was presented and visualized to fit the needs of the audience which are young people.

II. LITERATURE REVIEW

Data Analysis is the process of bringing order and structure to collected data. It turns data into information teams can use. Analysis is done using systematic methods to look for trends, groupings, or other relationships between different types of data.

Data visualization is the process of putting data into a chart, graph, or other visual format that helps inform analysis and interpretation. Data visuals present the analyzed data in ways that are accessible to and engage different stakeholders. Data visuals are also used to communicate results to meet key stakeholder needs. Multiple visuals will likely be needed to understand the larger change process and inform data use [5].

An information repository is a collection of interrelated information maintained across a network on multiple servers. It creates a unified resource for anyone connected with the system to access when they need information [6].

III. DATA GATHERING METHODOLOGY

A. COVID-19 Data Processing

In the Philippines, there is only one data source that all news agencies rely on for information and that is the Philippine Department of Health (DOH).

TABLE I. SUMMARY OF PUBLICLY AVAILABLE DATA SOURCES FOR TRACKING COVID-19 IN THE PHILIPPINES

Dataset	Open Access	Update Frequency	Timing	Scope	Data Reported
DOH Bulletin [7]	Yes	Daily	4PM, same day	National	Cases, Deaths, Recovery
DOH Situationer [7]	Yes	Daily	8PM, same day	National	Cases, Deaths, Recovery
DOH Data Drop [8]	Yes	Daily	10AM, next day	City	Cases, Deaths, Recovery, Testing Data, Bed Capacity

Case, death, recovery as well as testing data are downloaded from the sources such as the DOH Data Drop (<http://bit.ly/DataDropPH>) and undergo data transformation before loading to the dashboard's database.

B. Local and International News Processing

The purpose of this dashboard can be subdivided into three key points:

- **Accurate:** factual, correct info from reputable sources
- **Consolidated:** easy/convenient to navigate, salient points are combined already, simplified and streamlined
- **Easily digestible:** presented clearly for lay people to understand, not overly technical or complex

This is realized through our Multi-level Quality Assurance Flow.

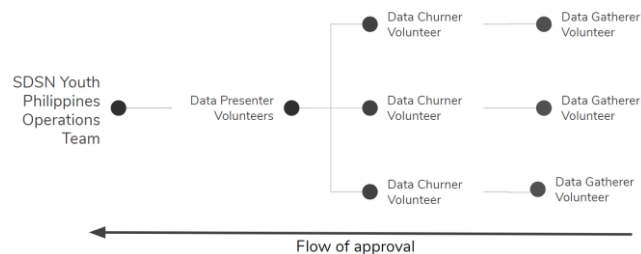


Fig. 1. Multi-level Quality Assurance Flow

The data gatherers will collate and consolidate all relevant news articles from government sources, news, online research, etc. They will pass on the data that they find to data churners for

fact checking, quality assurance, verification, etc. Once approved, they will pass on verified articles to data presenters for another round of fact checking and verification. Materials from presenters will be passed onto SDSN Youth team for posting on the site.

The purpose is to make sure no unverified data ends up on the site, ensure that each piece of information posted is at least triple checked.

Each team has its own flowchart or a four/three-way test to ensure that their submission is accurate, legitimate and factual.

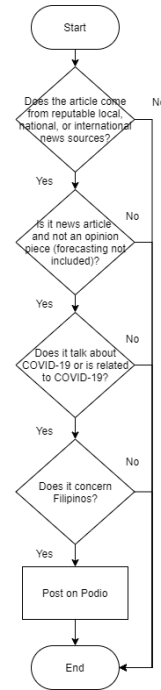


Fig. 2. Data Gatherer's Four-Way Test

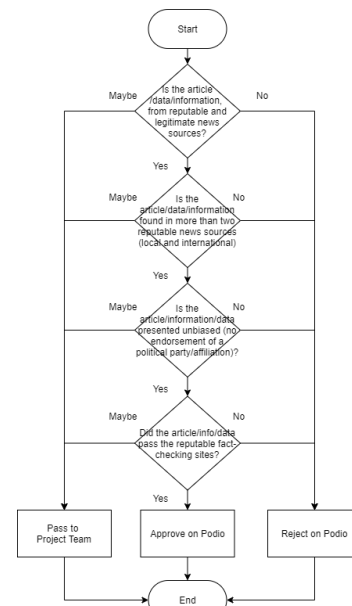


Fig. 3. Data Churner's Four-Way Test

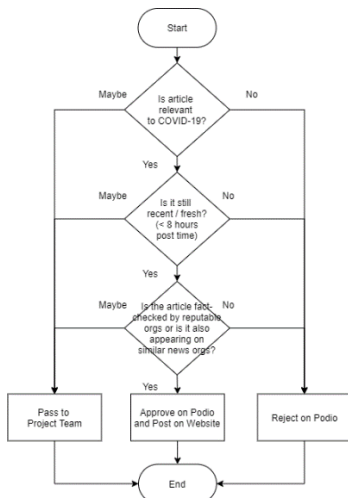


Fig. 4. Data Presenter's Three-Way Test

The teams use Podio to post and approve news sources and data before loading on to the database.

IV. DEVELOPMENT METHODOLOGY

Development of this Interactive Dashboard were days of prototype testing. This follows a near-Agile development approach since development man days are limited due to the rapid developments and increases in COVID-19 cases.

A. Analysis and Design

Presentation of the data is important in a dashboard. Thus, our dashboard design heavily relied on what the user wants to see in our dashboard such as the current accumulated count of COVID-19 confirmed cases, deaths, and recoveries. Also included are the daily case, death and recovery count and its trend (either upward, downward or steady). Other details such as Active Cases (or those that are not currently dead nor recovered), Case Fatality Rate and Case Recovery Rate, and the Overseas Filipino Worker and Health Worker Cases and Deaths.

The dashboard also features a carousel of important news and information about COVID-19 together with graphs of (1) cumulative data trend of confirmed cases, deaths, and recoveries, daily confirmed cases, deaths, and recoveries as well as the bar chart of female and male cases arranged by age group and a pie chart of health status (such as asymptomatic cases, critical cases, dead cases, mild cases, recovered cases, and sever cases).

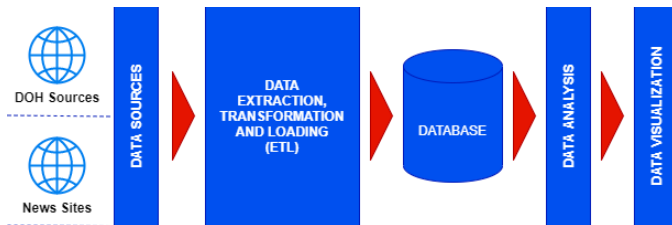


Fig. 5. Conceptual Framework

B. Development and Testing

The dashboard was created using PHP and Bootstrap with MySQL as its database. The production server is hosted by the UN SDSN Youth Philippines through its service provider.

Testing was done immediately after each sprint.

V. RESULTS AND DISCUSSION

All development work was done in sync with the needs and requirements as the action was happening. Thus, development is fast-paced, and the dashboard keeps on changing as the months progressed.

The result of the development and testing for the dashboard, both for the data visualization and data analysis had been impressive. As there were many users accessing the site.

Below are some screenshots of the Interactive Dashboard [9].

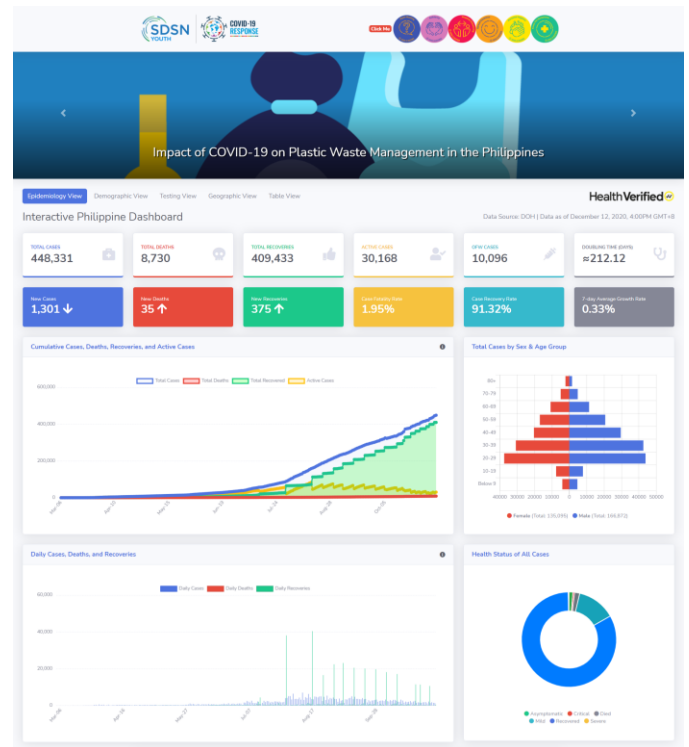


Fig. 6. Epidemiology View (Home Page)



Fig. 7. Testing View

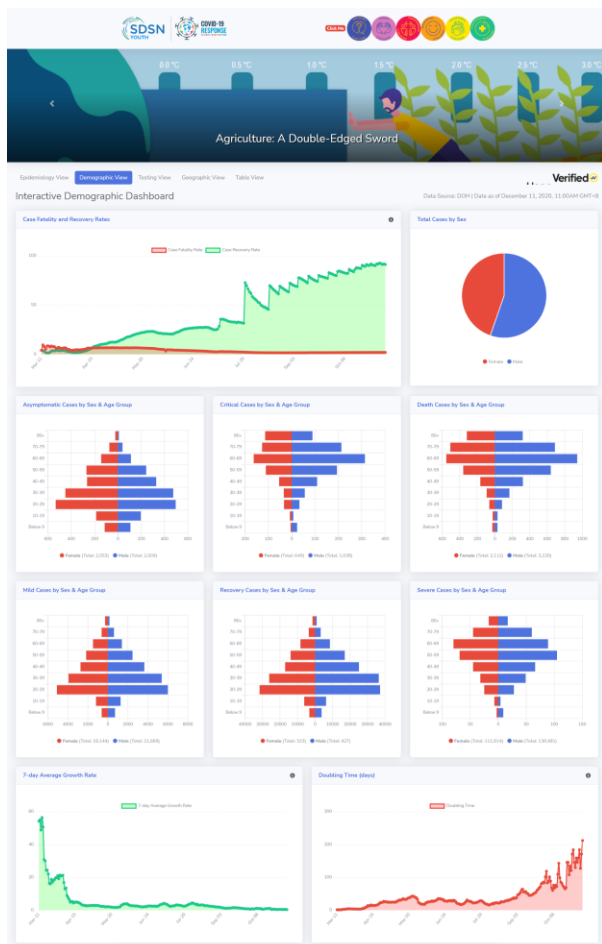


Fig. 8. Demographic View

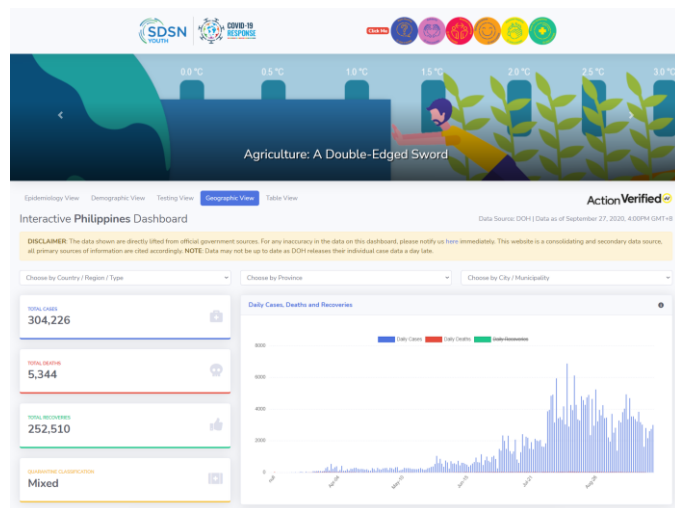


Fig. 9. Geographic View



Fig. 10. Consolidated Philippine News on COVID-19



Fig. 11. Demographic Vulnerability (Risk) Map

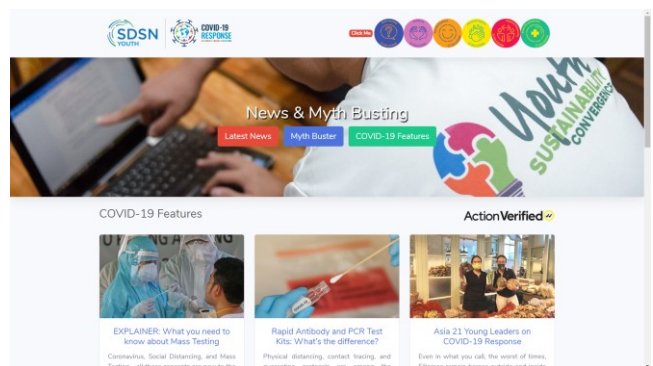


Fig. 12. COVID-19 Features (In-House Articles)



Fig. 13. Local and International Good News



Fig. 14. Personal Hygiene Tips

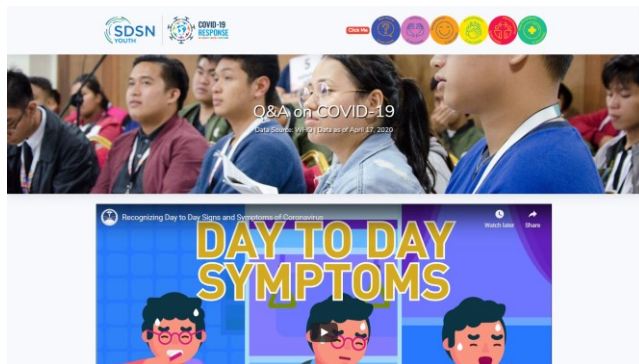


Fig. 15. Frequently Asked Questions on COVID-19

VI. CONCLUSION

Using the Interactive Dashboard, users found it easy to digest COVID-19 data and were updated in real time. The objectives were met where young people were the ones visiting the site and getting the needed information on COVID-19.

Based on the visitor data of the website, we have reached more than one hundred thousand (100,000) visitors on our site and averaging of around five hundred to one thousand (500-1,000) visitors per day.

VII. RECOMMENDATION

The researcher would like to recommend including more data that is not publicly available. The researcher also recommends further studies and use case for this system especially for the newly found variant/s of the Corona Virus Disease 2019.

ACKNOWLEDGMENT

Our thanks to the hardworking volunteers, staff, project implementation team, and the project management team of the SDSN Youth Philippines headed by Xavier Peredo for keeping the created dashboard running in tip top shape. Our deepest gratitude as well to the donors who believed in our cause.

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